Digital Electronics: Detailed Notes

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1. Introduction to Digital Systems

Digital System: A system that processes discrete values (usually binary: 0 and 1).

- Applications: Computers, calculators, digital watches, mobile phones, etc.
- Advantages over Analog: Noise immunity, scalability, accuracy, reliability.

2. Number Systems and Conversions

a. Types of Number Systems

- Binary (Base 2): 0, 1
- Octal (Base 8): 0 to 7
- Decimal (Base 10): 0 to 9
- Hexadecimal (Base 16): 0 to 9, A (10) to F (15)

b. Conversions

i. Decimal to Binary:

- Divide by 2 repeatedly and record remainders.
- E.g., 13 -> 1101

ii. Binary to Decimal:

- Multiply each bit with 2^position and add.
- E.g., 1101 = 1x2^3 + 1x2^2 + 0x2^1 + 1x2^0 = 13

iii. Binary to Octal/Hex:

- Group 3/4 bits from LSB.
- Binary 101101 -> Octal = 55

... (truncated for space)